CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 3: Squares and Square Roots

Topic-wise short notes with solved examples

Perfect squares and their properties, and finding square roots by prime factorisation and by the long division method, including square roots of decimals and fractions.

1 Squares

- The **square** of a number is the number multiplied by itself: $n^2 = n \times n$.
- A natural number that is the square of a natural number is a **perfect square**: 1, 4, 9, 16, 25, ...
- In the prime factorisation of a perfect square, every prime occurs an **even** number of times.

n	11	12	13	14	15	16	17	18	19	20	25
n^2	121	144	169	196	225	256	289	324	361	400	625

2 Properties of perfect squares

- A perfect square ends in **0, 1, 4, 5, 6 or 9**. A number ending in 2, 3, 7 or 8 is never a perfect square.
- A perfect square ends in an **even number of zeros**.
- The square of an even number is even; the square of an odd number is odd.
- A perfect square leaves remainder 0 or 1 on division by 3, and remainder 0 or 1 on division by 4.
- **Sum of the first n odd numbers** $= n^2$.
- Between n^2 and $(n+1)^2$ there are $2n$ natural numbers.
- $(n+1)^2 - n^2 = 2n + 1$.
- **Pythagorean triplet**: for any natural number $m > 1$, the numbers $2m$, $m^2 - 1$, $m^2 + 1$ form a triplet.

Example 2.1 Which of these cannot be perfect squares: 2457, 5776, 6400, 9000?

SOLUTION

2457 ends in 7. 9000 ends in an odd number of zeros.

2457 and 9000

Example 2.2 Without adding, find $1 + 3 + 5 + \ldots + 19$.

SOLUTION

There are 10 odd numbers.

$10^2 = 100$

Example 2.3 Write a Pythagorean triplet one of whose members is 16.

SOLUTION

$2m = 16$ gives $m = 8$: $m^2 - 1 = 63$, $m^2 + 1 = 65$.

16, 63, 65

3 Quick squares

- Use identities: $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$.
- A number ending in 5: $(10a + 5)^2 = 100a(a + 1) + 25$. Write $a(a + 1)$ followed by 25.

Example 3.1 Find 43^2, 98^2 and 65^2.

SOLUTION

$43^2 = 1600 + 240 + 9$; $98^2 = 10000 - 400 + 4$; 65^2: $6 \times 7 = 42$, then 25.

1849, 9604 and 4225

4 Square root by prime factorisation

- If $x^2 = y$, then x is a **square root** of y. The symbol $\sqrt{y}$ means the positive square root.
- **Method**: write the number as a product of primes; make **pairs** of equal primes; take one from each pair and multiply.
- If a prime is left unpaired, the number is not a perfect square. Multiplying or dividing by the unpaired primes makes it one.

Example 4.1 Find $\sqrt{7056}$.

SOLUTION

$7056 = 2^4 \times 3^2 \times 7^2$

$2^2 \times 3 \times 7 = 84$

Example 4.2 Find the smallest number by which 1800 must be multiplied to make it a perfect square, and the square root of the result.

SOLUTION

$1800 = 2^3 \times 3^2 \times 5^2$. One 2 is unpaired.

Multiply by 2; $\sqrt{3600} = 60$

Example 4.3 Find the smallest number by which 2925 must be divided to get a perfect square.

SOLUTION

$2925 = 3^2 \times 5^2 \times 13$

13; the quotient 225 has square root 15

5 Square root by long division

- Mark off the digits in **pairs (periods)** from the units place. For decimals, pair to the left and to the right of the decimal point.
- Find the largest number whose square is not more than the first period. It is the first digit of the root and the first divisor.
- Subtract, bring down the next period. **Double the quotient** so far and write it with a blank digit.
- Choose the largest digit for the blank so that (new divisor) $\times$ (that digit) is not more than the dividend. Repeat.
- The number of digits in the square root of an n-digit perfect square is $\frac{n}{2}$ if n is even and $\frac{n+1}{2}$ if n is odd.

Example 5.1 Find $\sqrt{2304}$ by long division.

SOLUTION

Periods: 23 | 04. $4^2 = 16$; $23 - 16 = 7$; bring down 04 to get 704.

Double 4 is 8. $88 \times 8 = 704$. Remainder 0.

48

Example 5.2 Find $\sqrt{54756}$.

SOLUTION

Periods: 5 | 47 | 56. $2^2 = 4$, remainder 1, dividend 147. $43 \times 3 = 129$, remainder 18, dividend 1856. $464 \times 4 = 1856$.

234

Example 5.3 Find the least number that must be subtracted from 5607 to make it a perfect square.

SOLUTION

Long division gives quotient 74 with remainder 131 ($74^2 = 5476$).

131

Example 5.4 Find the least number that must be added to 1300 to make it a perfect square.

SOLUTION

$36^2 = 1296 < 1300 < 37^2 = 1369$

69

6 Decimals and fractions

- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ and $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.
- For a decimal, the square root has **half as many** decimal places as the number.
- To find a square root correct to n decimal places, work to $n + 1$ places and round off.

Example 6.1 Find $\sqrt{\frac{441}{961}}$ and $\sqrt{5\frac{19}{25}}$.

SOLUTION

$\frac{21}{31}$; and $5\frac{19}{25} = \frac{144}{25}$ has square root $\frac{12}{5}$

$\frac{21}{31}$ and $2\frac{2}{5}$

Example 6.2 Find $\sqrt{12.25}$ and $\sqrt{0.0169}$.

SOLUTION

3.5 and 0.13

Example 6.3 Find $\sqrt{2}$ correct to two decimal places.

SOLUTION

Long division of 2.000000 gives 1.414...

1.41

7 Word problems

Example 7.1 The area of a square field is 5184 sq m. Find its side and perimeter.

SOLUTION

Side $= \sqrt{5184} = 72$ m

72 m; 288 m

Example 7.2 1225 students stand in rows so that the number of rows equals the number of students in each row. Find the number of rows.

SOLUTION

$\sqrt{1225} = 35$

8 Common mistakes

- Pairing digits from the left instead of from the units place.
- Forgetting to double the whole quotient at each step.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.
- Giving $\sqrt{0.9} = 0.3$. In fact $0.3^2 = 0.09$.
- Leaving an unpaired prime under the root in the factor method.
- Deciding that a number is a perfect square from its last digit alone.

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